Calculus with Parametric and Polar Curves

Section 1: Derivatives of Parametric Curves

1 Given $x(t) = t^3 - 3t$ and $y(t) = t^2 + 1$, find $\frac{dy}{dx}$ in terms of t.

a. $\frac{2t}{3t^2 - 3}$

b. $\frac{3t^2 - 3}{2t}$

c. $\frac{2t}{3t^2}$

d. $\frac{2}{3t}$

2 Find the equation of the tangent line to the parametric curve $x = 2\cos t$, $y = 3\sin t$ at $t = \pi/4$.

a. $y - \frac{3\sqrt{2}}{2} = -\frac{3}{2}(x - \sqrt{2})$

b. $y - \sqrt{2} = -\frac{2}{3}(x - \frac{3\sqrt{2}}{2})$

c. $y = -\frac{3}{2}x$

d. $y - 3 = -(x - 2)$

3 State the formula for the second derivative $\frac{d^2y}{dx^2}$ of a parametric curve.

a. $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left[\frac{dy}{dx}\right]}{\frac{dx}{dt}}$

b. $\frac{d^2y}{dx^2} = \frac{y''(t)}{x''(t)}$

c. $\frac{d^2y}{dx^2} = \frac{y''(t)}{[x'(t)]^2}$

d. $\frac{d^2y}{dx^2} = \frac{d}{dt}\left[\frac{dy}{dx}\right]$

Section 2: Arc Length of Parametric Curves

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4 State the formula for the arc length of a parametric curve from $t = a$ to $t = b$.

a. $L = \int_a^b \sqrt{[x'(t)]^2 + [y'(t)]^2}\, dt$

b. $L = \int_a^b (x'(t) + y'(t))\, dt$

c. $L = \int_a^b \sqrt{1 + [y'(t)]^2}\, dt$

d. $L = \int_a^b \sqrt{[x(t)]^2 + [y(t)]^2}\, dt$

5 Find the arc length of $x = 3\cos t, y = 3\sin t$ for $0 \le t \le 2\pi$.

a. 6π

b. 3π

c. 9π

d. 12π

6 Find the arc length of $x = t^2, y = \frac{2}{3}t^3$ from $t = 0$ to $t = 1$.

a. $\frac{2}{3}(2\sqrt{2} - 1)$

b. $\frac{1}{3}(2\sqrt{2} - 1)$

c. $\sqrt{2} - 1$

d. $2\sqrt{2}$

Section 3: Slopes and Tangents of Polar Curves

7 Given $r = f(\theta)$, how are the Cartesian coordinates related to polar coordinates?

a. $x = r\cos\theta, y = r\sin\theta$

b. $x = r\sin\theta, y = r\cos\theta$

c. $r = x + y, \theta = y/x$

d. $x = r^2\cos\theta, y = r^2\sin\theta$

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8 Find the slope $\frac{dy}{dx}$ of the polar curve $r = 2$ at $\theta = \pi/4$.

a. -1

b. 1

c. 0

d. Undefined

9 At what angles $\theta \in [0, \pi)$ does the cardioid $r = 1 + \cos\theta$ pass through the pole ($r = 0$)?

a. $\theta = \pi$

b. $\theta = \pi/2$

c. $\theta = 0$

d. $\theta = 3\pi/4$

Section 4: Area in Polar Coordinates

10 State the formula for the area enclosed by a polar curve $r = f(\theta)$ from $\theta = \alpha$ to $\theta = \beta$.

a. $A = \frac{1}{2}\int_{\alpha}^{\beta} [f(\theta)]^2 \, d\theta$

b. $A = \int_{\alpha}^{\beta} f(\theta) \, d\theta$

c. $A = \pi\int_{\alpha}^{\beta} [f(\theta)]^2 \, d\theta$

d. $A = \frac{1}{2}\int_{\alpha}^{\beta} f(\theta) \, d\theta$

11 Find the area enclosed by one petal of the four-petaled rose $r = 2\cos(2\theta)$.

a. $\frac{\pi}{2}$

b. π

c. 2π

d. $\frac{\pi}{4}$

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12 Find the area of the region inside the circle $r = 4\sin\theta$.

a. 4π

b. 16π

c. 2π

d. 8π

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